Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *HOPE_DAC* | ***Context:*** *Global NLP/AI Research Community — Computational Mental Health (HOPE Benchmark Assessment)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Performance is reported via macro-F1, weighted-F1, and accuracy [Q63], with label-wise F1 [Q76] and 3-fold cross-validation [Q87]. SPARTA-TAA achieves 60.29 macro-F1 / 64.53 weighted-F1 / 64.75% accuracy [Q73, Q85] with statistical significance vs. CASA [Q88]. Strengths include label-wise reporting that surfaces ACK weakness (F1=47.86%) [Q77] and proper statistical testing. However, the OF dimension (MODERATE priority) faces a confirmed mismatch: evaluation is single-label classification only [Q63], while the elicited downstream chatbot use case benefits from multi-label or ranked outputs (A3 explicitly confirms 'multi-label or ranked outputs would yield more contextually appropriate downstream responses'). The benchmark itself acknowledges multi-intent utterances [Q30] but defers the form change as future work. The emotion annotation experiment was abandoned due to imbalance [Q126–Q129], leaving no affective dimension in the output form. No subgroup-disaggregated performance metrics are reported, which becomes a regulatory issue under emerging FDA equity expectations [WEB-20, WEB-27]. Confusion analysis [Q90, Q91, Q92] is reported but not disaggregated by speaker role or cultural subgroup.

ID	Evidence
Q30	“Sometimes, an utterance can have multiple dialog-acts; however, they are rare in the annotated dataset. Hence, for simplicity, we consider only one (primary) label for each utterance.” (p.3)
Q63	“To measure the performances of SPARTA and other baseline systems, we compute macro-F1, weighted-F1, and accuracy scores.” (p.6)
Q73	“Since SPARTA incorporates three major components – local context, global context, and speaker-aware modules. The SPARTA-TAA system obtains the best scores of 60.29 macro-F1, 64.53 weighted-F1,” (p.6)
Q76	“We also present the label-wise performance of SPARTA in Table 4.” (p.7)
Q77	“We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%.” (p.7)
Q85	“In comparison, SPARTA-TAA obtains significant improvements over all baselines. It reports improvements of +8.64%, +8.58%, and +6.29% in macro-F1 (60.29), weighted-F1 (64.53), and accuracy (64.75%), respectively, as compared to CASA suggesting the incorporation of local context extremely effective.” (p.7)
Q87	“Moreover, we also report the mean of the 3-fold cross-validation results for both SPARTA-MHA and SPARTA-TAA, and the results are consistent with the train-val-test split case.” (p.7)
Q88	“We also perform statistical significance T-test comparing SPARTA-TAA and the best performing baseline (CASA). We observe that our results are significant with > 95% confidence across macro-F1 (p-value= 0.009), weighted-F1 (p-value= 0.014), and accuracy (p-value= 0.048) values.” (p.7)
Q90	“Quantitative analysis: We report the confusion matrix for SPARTA-TAA in Figure 4.” (p.7)
Q91	“We observe three pairs with significant error-rates (≥ 25%) – YNQ:IRQ (26%), OD:ID (43%), ID:ACK (28%).” (p.7)
Q92	“For the prediction of information delivery (ID), SPARTA is confused most of the time with other classes – 19% with PA, 20% with NA, 43% with OD.” (p.7)
Q126	“In earlier phases of experiments, we also considered the role of emotion in dialogue act classification.” (p.11)
Q127	“We annotated a subset of our dataset with three emotion classes consisting of 'positive', 'negative' and 'neutral'.” (p.11)
Q128	“We found that around ~ 70% of utterances by the patients belonged to 'negative' class whereas ~ 90% of therapists' utterances belonged to 'neutral' class.” (p.11)
Q129	“Due to such imbalance, this data was not used in the final version of our proposed architecture.” (p.11)

WEB-20	Orrick (Nov 2025): FDA DHAC signals sponsors will need 'evidence of equitable performance across populations and languages' for AI mental health devices (https://www.orrick.com/en/Insights/2025/11/FDAs-Digital-Health-Advisory-Committee-Considers-Generative-AI-Therapy-Chatbots-for-Depression)
WEB-27	FDA document on generative AI in digital mental health medical devices (https://www.fda.gov/media/189391/download)

Strengths

- Label-wise F1 reporting [Q76, Q77] surfaces uneven benchmark signal across DA categories.
- Statistical significance testing with t-tests at >95% confidence [Q88] follows good methodological practice.
- 3-fold cross-validation results consistent with primary split [Q87] indicates result stability.
- Confusion matrix analysis [Q90, Q91, Q92] provides interpretable error structure information.

ID	Evidence
Q76	“We also present the label-wise performance of SPARTA in Table 4.” (p.7)
Q77	“We can observe that SPARTA consistently yields good scores for the majority of the dialogue-acts, except for the Acknowledgement (ACK) where it records F1-score of merely 47.86%.” (p.7)
Q87	“Moreover, we also report the mean of the 3-fold cross-validation results for both SPARTA-MHA and SPARTA-TAA, and the results are consistent with the train-val-test split case.” (p.7)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality is single categorical label per utterance [Q63]. The elicited downstream chatbot use case benefits from multi-label or ranked outputs (A3) — this is a documented mismatch.

OF-2: Not applicable — the use case is text-only, no TTS requirement applies.

OF-3: Not applicable — the user population is researchers (English-proficient), not literacy-constrained end users. Downstream chatbot literacy concerns are out of scope per the regional context's IF=LOWER framing.

OF-4: Documented form mismatches: (a) single-label vs. multi-label/ranked output need [Q30, Q63, A3]; (b) no affective dimension after emotion annotation was abandoned [Q129]; (c) no subgroup-disaggregated metrics, which conflicts with emerging FDA equity expectations [WEB-20, WEB-27]; (d) macro-F1 weights all classes equally despite label imbalance and may obscure systematic culture-related failures.

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WEB-27	FDA document on generative AI in digital mental health medical devices (https://www.fda.gov/media/189391/download)

Information Gaps

- No subgroup-disaggregated performance metrics — would require running HOPE-trained classifiers on stratified evaluation samples.

Remediation

Gap 1: No subgroup-disaggregated performance metrics are reported, conflicting with emerging FDA equity expectations [WEB-20, WEB-27].

Recommendation: Adopt a reporting standard that disaggregates macro-F1, weighted-F1, and per-label F1 by speaker role and (when subgroup data are available) by demographic subgroup. Add ranked-output metrics (top-k accuracy, mean reciprocal rank) to align with multi-intent downstream needs.

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